

Symbolic rules and anchoring improve the search

~80×

smaller concrete rule sets
with full prior-rule coverage

~11.6×

Symbolic Synthesis
speedup from
property grouping

Full QSymb (Concrete rule + Anchored rule + Canonical rule) versus ablations

Concrete only

78%

100%

97%

as good or better

Concrete + Canonical symbolic

31%

97%

as good or better

■ strictly better

■ equal

Takeaway

- **Expressive and effective optimization**: symbolic rules capture long subcircuits with superposition; anchoring enables gate reductions, overcoming limited reach.
- **Compact rules with provable coverage**: non-derivable concrete rules and canonical symbolic rules control rule explosion, with completeness within size and qubit bounds.

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Github: github.com/VeriGu/QSymb

QSymb · GitHub

